# Chi-Square Analysis

Stroke Status & Hospital Readmission Rates

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## Research Question

Is there a relationship between experiencing a stroke and rate of readmission?

## Stakeholder Benefit

If patients who experience a stroke are being readmitted more often than average, the hospital should adjust by asking stroke patients to stay longer for their safety.

## Methodology

I have chosen to use chi square analysis because this will allow us to mathematically determine the strength of the relationship between the variables 'stroke' and 'readmission'. The categorical readmission and stroke columns will be the required data for this chi square analysis.

## Analysis Output

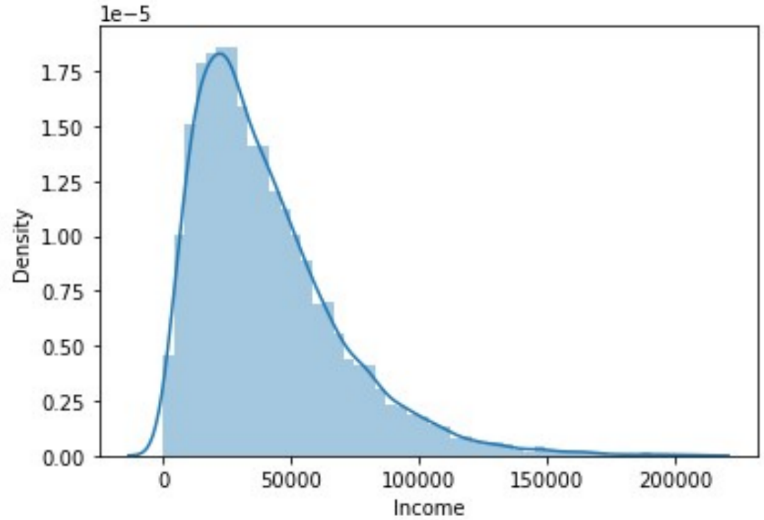
```
In [36]: runfile('C:/Users/velez/OneDrive/Desktop/
School/D207/Brandon Velez D207 PA.py', wdir='C:/Users/
velez/OneDrive/Desktop/School/D207')
   CaseOrder Customer_id  ... Item7 Item8
0          1     C412403  ...     3     4
1          2     Z919181  ...     3     3
2          3     F995323  ...     3     3
3          4     A879973  ...     5     5
4          5     C544523  ...     4     3

[5 rows x 50 columns]
Stroke     No   Yes
ReAdmis
No       5071  1260
Yes      2936   733
Stroke         No       Yes
ReAdmis
No       0.800979  0.199021
Yes      0.800218  0.199782
0.9474770077616069
p value is 0.9474770077616069
Independent (accept H0)
```

3. Justify why you chose this analysis technique.

*Chi-square output: contingency table, expected proportions, p-value*

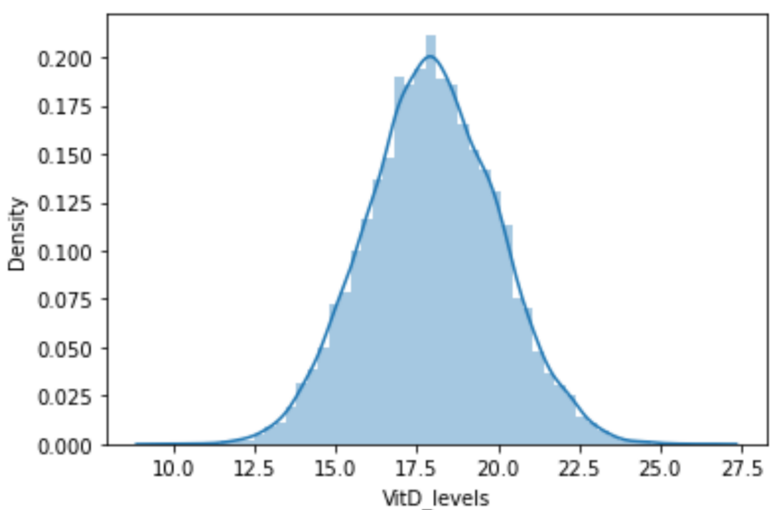
## Univariate Analysis




The population density graph is heavily right skewed, as there is an extremely high peak ir

*Income density distribution*

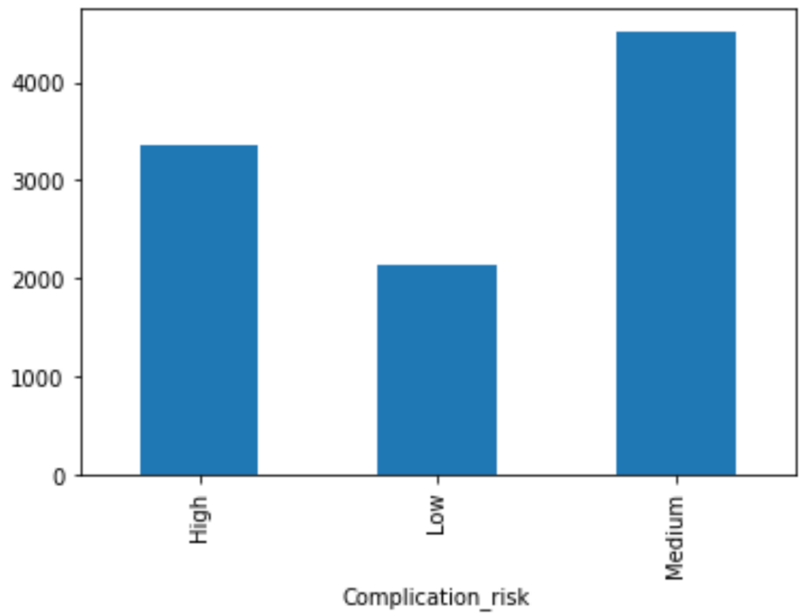
The income density graph is right skewed, meaning there is a high peak in the lower values on the left, and then has a very long tail as values increase to the right.




Vitamin D levels look to be normally distributed. Here is the .describe output for the vitD_
column.

*Vitamin D levels distribution*

The population density graph is heavily right skewed, as there is an extremely high peak in density early on, and then tails off for quite a long time as the population values increase and density decreases. Vitamin D levels look to be normally distributed.




The categorical complication risk distribution chart is a discrete non-uniform distribution.

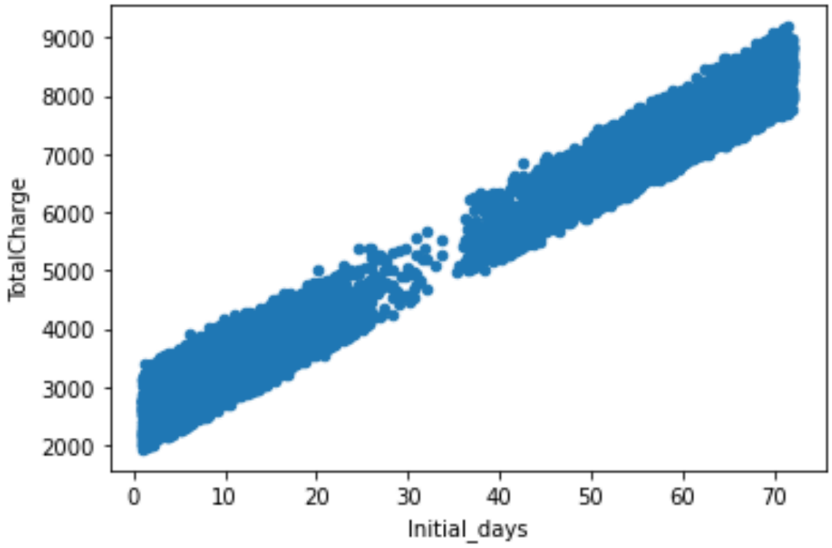
*Complication risk and services rendered*

The categorical complication risk distribution chart is a discrete non-uniform distribution. The categorical services chart follows a non-uniform distribution as well.

## Bivariate Analysis

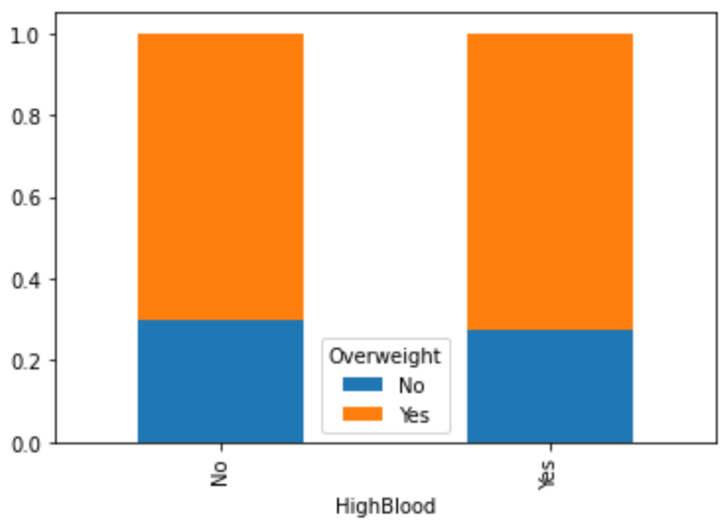
D. In a document file, identify the distribution of **two** continuous variables and **two** categ
variables using bivariate statistics from your cleaned and prepared data.
1. Represent your findings in part D visually as part of your submission.

The initial days to total charge visualization follows a linear distribution, meaning moı
will correlate very strongly to a higher total charge. This is not perfect however, as
value of initial days can correlate to multiple total charge values.




*Initial days vs. total charge*

The initial days to total charge visualization follows a linear distribution, meaning more days will correlate very strongly to a higher total charge. This is not perfect however, as any value of initial days can correlate to multiple total charge values.




*Overweight status vs. high blood pressure*

Overweight to high blood pressure is a discrete uniform distribution, meaning there is no significant difference in blood pressure classification when comparing overweight to non overweight patients.

## Results

The Null hypothesis has proven to be true, there is no statistical relationship between rate of readmission and stroke status. Based on the results of the hypothesis test, no action is needed on the hospital's part to adjust how they are handling patients who have experienced a stroke. There is no relationship between experiencing a stroke and being readmitted to the hospital at a higher rate.

## Limitations

A limitation of this data analysis would be that this data is only coming from a singular hospital location over a short period of time. While these results are valid in the context of the provided data, this is too narrow in scope to make any general assumptions on the relationship between stroke status and readmission rate. If thousands of hospitals were sampled across the world over a period of multiple decades, the data analysis would be resistant to any outlying values or geographical differences, and could therefore be generalized and considered more statistically robust.

There will always be the issue of self-reporting bias, as patients could be incorrect, write illegibly on intake forms, or just lie. The hospital provided fields are subject to human error, but likely much less so than patient reported information as trained staff are inputting that data.

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Tools: Python, pandas, scipy.stats, seaborn, matplotlib